Financial Market Vision & Quantitative Pattern Detection

Interactive notebook demonstrating:

1. Fetching real market data
2. Quantitative candlestick pattern detection
3. Visualization of detected patterns
4. Basic computer-vision analysis of chart images

1. Imports & Setup

In [ ]:
```
import sys
sys.path.append("..")

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from src.data_fetcher import fetch_ohlcv
from src.pattern_detector import detect_all_patterns, summarize_patterns
from src.visualizer import plot_candles_with_patterns, plot_pattern_counts
from src.vision_analyzer import analyze_chart_image, create_synthetic_candle_ima

sns.set_style("whitegrid")
plt.rcParams["figure.figsize"] = (12, 6)
```

2. Fetch Market Data

In [ ]:
```
TICKER = "AAPL"
PERIOD = "6mo"

df = fetch_ohlcv(ticker=TICKER, period=PERIOD, interval="1d", save=True)
df.tail()
```

In [ ]:
```
print(df.describe())
df[["Close"]].plot(title=f"{TICKER} Close Price")
plt.show()
```

3. Quantitative Pattern Detection

In [ ]:
```
pattern_df = detect_all_patterns(df)
print("Pattern signal matrix (last 10 rows):")
pattern_df.tail(10)
```

In [ ]:
```
events = summarize_patterns(pattern_df)
print(f"Total pattern occurrences: {len(events)}\n")
print("Most recent detections:")
```

```
for dt, name in events[-20:]:
    print(f"  {dt.date()}  →  {name}")
```

In [ ]:
```
plot_pattern_counts(pattern_df)
plt.show()
```

4. Visualize Price Chart with Patterns

In [ ]:
```
plot_candles_with_patterns(
    df,
    events,
    title=f"{TICKER} – Detected Candlestick Patterns ({PERIOD})",
    save_path=f"../charts/{TICKER}_{PERIOD}_notebook_chart.png",
)
# The function saves and closes; re-plot interactively if desired
print("Chart saved. Re-running a quick view...")

# Quick interactive view of recent data
recent = df.tail(60)
recent_events = [(d, n) for d, n in events if d in recent.index]
plot_candles_with_patterns(
    recent,
    recent_events,
    title=f"{TICKER} – Last 60 days",
)
plt.show()
```

5. Computer Vision Analysis of Chart Image

In [ ]:
```
# Create a synthetic candlestick image from recent data
ohlc = list(zip(
    recent["Open"].tolist(),
    recent["High"].tolist(),
    recent["Low"].tolist(),
    recent["Close"].tolist(),
))
synth_path = f"../charts/{TICKER}_synth.png"
create_synthetic_candle_image(ohlc, save_path=synth_path)
print(f"Synthetic chart saved to {synth_path}")
```

In [ ]:
```
result = analyze_chart_image(
    synth_path,
    output_path=f"../charts/{TICKER}_vision_annotated.png",
)
print("Vision pipeline results:")
for k, v in result.items():
    if k != "contour_areas":
        print(f"  {k}: {v}")
print(f"  (showing first few contour areas): {result['contour_areas'][:5]}")
```

6. Next Steps / Ideas

- Add more patterns (Harami, Piercing, Dark Cloud, Tweezer tops/bottoms…)
- Measure statistical edge of each pattern (forward returns)

- Train a CNN on labeled chart screenshots for pure vision-based detection
- Real-time scanning across a universe of tickers
- Streamlit dashboard for interactive exploration